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LAB1 RAPPORT

Forward and Backward Chaining for an Expert Agent

Objective:

Implement forward chaining and backward chaining mechanisms in a rule-based expert system. The goal is to equip an agent with logical reasoning capabilities.

1. Knowledge representation:

- A rule class represents a rule with:
 - p: premises (conditions).
 - c: conclusion.
- A knowledge base: a set of initially known facts.
- A goal to prove.

2. Forward Chaining Algorithm:

The **chain function** implements forward chaining by iteratively applying rules to expand the fact base. It loops through all rules, checking if all premises exist in facts. If satisfied, it adds new conclusions, marking **changed = true** to continue iteration until no new facts are derived. Success is determined by checking if all goals are in the final fact base.

3. Backward Chaining Algorithm:

The **back_ch function** implements backward chaining by recursively proving goals from the first (**fp.get(0)**) using the rule base (**bdr**) and fact base (**bdp**). If the goal is provable via **takegoal**, it adds the goal to **bdp** and recurses on remaining goals. Empty goal list yields success.

4. Interface Implementation:

The interface visualizes the inference process through real-time fact base updates and rule firing traces. It allows the user to enter an initial fact base and a goal to be proven. After launching the reasoning process, the system displays the sequence of rules explored during the proof search, as well as the result indicating whether the goal has been reached.

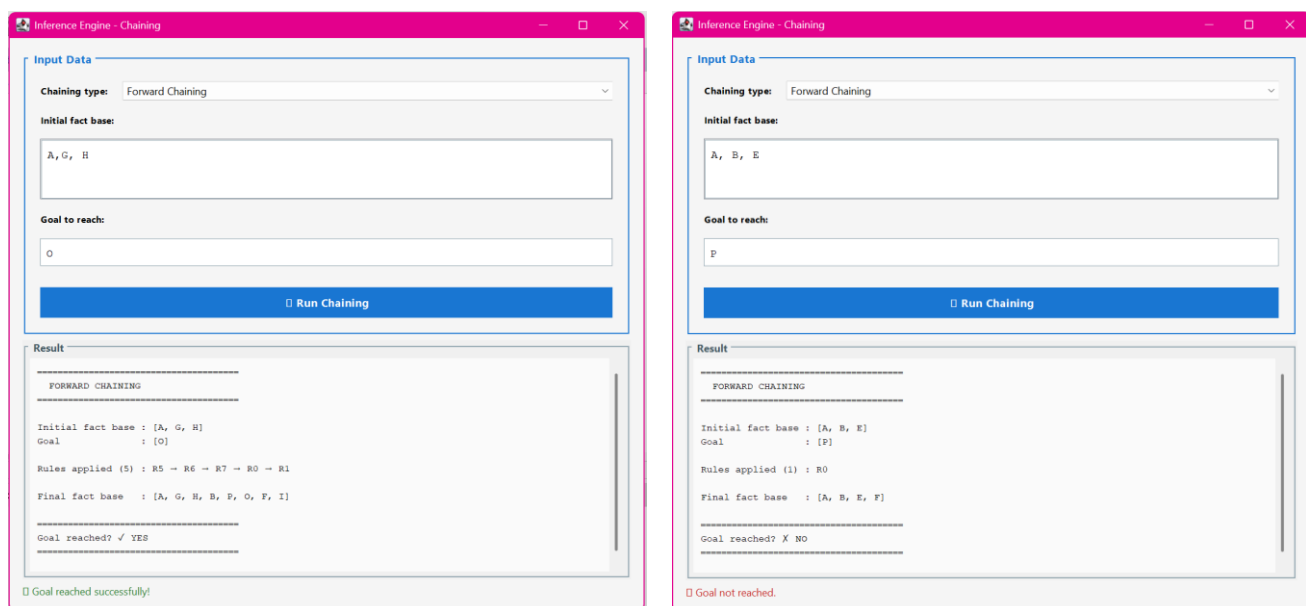


Figure 1– the interface of forward chaining reasoning

2. Conclusion

This lab implemented the basic codes of forward and backward chaining reasoning and then the inference engine for real-time visualization.